

1. In recent weeks many anti-war protests have taken place at different university campuses in North America. A survey conducted last April by the Angus Reid Institute asked Canadians if they believe it is acceptable to protest in university campuses. Here is the data for Québec, Ontario, and Alberta:

	Acceptable	Never Acceptable	Not sure/Can't say	Total
Québec	342	60	25	427
Ontario	398	68	19	485
Alberta	180	19	8	207
Total	920	147	52	1119

- (a) (2 points) Find the probability that a randomly selected person in this sample believes it is acceptable to protest at university campuses.
- (b) (2 points) Find the probability that a randomly selected person in this sample is from Québec, given that they believe it is acceptable to protest at university campuses.
- (c) (2 points) Find the probability that a randomly selected person in this sample is from Ontario and believes it is acceptable to protest at university campuses.
- (d) (2 points) Are the events “believes it is acceptable” and “from Alberta” independent? Justify your answer mathematically.
2. AleaLand’s parliament has 20 members: 8 are liberal, 7 are conservative, and 5 are independent. They will randomly select 5 of the members to form a committee.
- (a) (3 points) What is the probability that the committee has only liberals?
- (b) (3 points) What is the probability that the committee has no conservatives?
3. In a telephone survey conducted on a sample of Canadian adults, the respondents were asked their opinion on the current cost of university education. The results have shown that 60% of the respondents said *too high*, 28% said *the right amount*, and the remaining percentage answered *too little*.
Among those who responded *too high*, 58% have a child currently studying in university. For those who said *the right amount*, 30% have a child in university. Finally, among those who answered *too little*, only 8% have a child currently in university.
- (a) (3 points) Among the parents with a child in university, what is the percentage who believe the cost of university is *too high*?
- (b) (3 points) Among the parents without a child in university, what is the percentage who believe the cost of university is *too high*?
4. According to the 2011 Census conducted by Statistics Canada, the distribution of household sizes in Canada are as follows:

number of persons	1	2	3	4	5
percentage	27.6%	34.1%	15.6%	14.3%	8.4%

Let X be the number of persons living in a randomly selected household (in 2011).

- (a) (2 points) Calculate the probability that a randomly selected household (in 2011) had 4 or more persons.
- (b) (2 points) Calculate the expected value of X .
- (c) (3 points) Calculate the standard deviation of X .

5. Felicia only drinks coffee occasionally. On any given day, she will drink coffee 19% of the time.
- (a) (3 points) Find the probability that, in the next 2 weeks (14 days), Felicia will drink coffee on exactly 4 of those days.
 - (b) (5 points) Find the probability that, in the next 2 weeks (14 days), Felicia will drink coffee on 2 or more days.
6. Mature apple trees yield, on average, 600 pounds of apples, with a standard deviation of 110 pounds. Assume that the yield of a mature apple tree is normally distributed.
- (a) (4 points) What is the probability that a randomly selected mature apple tree will produce between 480 lbs and 816 lbs of apples?
 - (b) (5 points) The apple producer's association recommends specific trimming methods to stimulate the production of low-yield apple trees. If this trimming method is to be applied to the apple trees with the lowest 4% yield, what production level does this correspond to?
 - (c) (5 points) In a small orchard of 24 mature apple trees, find the probability that the average yield of those 24 trees will be larger than 672 lb.
7. Houses built before 1960 are likely to contain asbestos in their construction materials. Previous estimates from examining houses built in Quebec reveal that 21.3% of houses built before 1960 contain asbestos. A pilot government program is mandated to determine the costs of asbestos removal in a random sample of 450 houses built before 1960 selected across the province. Let X be the number of houses built before 1960 in this sample that contain asbestos.
- (a) (2 points) What is the expected value of X ?
 - (b) (2 points) What is the standard deviation of X ?
 - (c) (5 points) Find the probability that at least 90 houses (inclusively) and less than 120 houses (exclusively) in this sample contain asbestos. Use a normal approximation.
8. A team of astronomers set out to study the celestial bodies. They observed a sample of 26 stars and measured their luminosity. If the sample mean luminosity is found to be 8.2 with a sample standard deviation of luminosity 2.5.
- (a) (4 points) What is the 99% confidence interval for the true mean luminosity of stars in the population, assuming that the star luminosity follows a normal distribution?
 - (b) (1 point) Interpret your confidence interval from part (a).
9. A research agency is commissioned to gauge the prevalence of smartphone ownership among adult residents. They surveyed 500 adults from various neighborhoods across the city of Montreal. Of these 500 individuals surveyed, 421 responded affirmatively, indicating they own smartphones.
- (a) (4 points) What is the 95% confidence interval for the true proportion of adult residents in this city who possess smartphones?
 - (b) (1 point) Interpret your confidence interval from part (a).
10. (3 points) A health agency wants to construct a 95% confidence interval for the mean height of a 5-year-old. The standard deviation of a 5-year-old is known to be 1.5 inches. What is the minimum sample size required to ensure the margin of error is within 0.5 inches?

11. Arsenic occurs naturally in some groundwater. A mean arsenic level of 8.0 parts per billion (ppb) is considered safe for agricultural use. A well in Texas is used to water cotton crops. This well is tested regularly for arsenic. A random sample of 36 tests gave a sample mean of 7.2 ppb arsenic and a standard deviation of 1.9 ppb. Use the data to test the claim that the average level of arsenic in this well is less than 8 ppb. Use a 2% significance level.

(a) (2 points) State the null and alternative hypotheses.

- H_0 :
- H_a :

(b) (3 points) Calculate the test statistic.

(c) (2 points) Find (or estimate) the p -value for the test.

(d) (2 points) *State* and *interpret* your conclusion.

12. A random sample of 228 registered voters in Ontario showed that 141 voted in the last general election. A random sample of 216 registered voters in Quebec showed that 125 voted in the most recent general election. Do these data indicate that the population proportion of voter turnout in Ontario is higher than that in Quebec? Use a 5% significance level.

(a) (2 points) State the null and alternative hypotheses.

- H_0 :
- H_a :

(b) (4 points) Calculate the test statistic.

(c) (2 points) Find (or estimate) the p -value for the test.

(d) (1 point) State your conclusion.

13. A teacher is developing a new teaching method and wants to assess its impact on students' test scores. To evaluate this, the teacher gathers data from the same group of students before and after exposure to the new method, resulting in the following table:

Students	Old Method	New Method
Amir	75	78
Blake	80	82
Clara	70	67
Demi	85	87
Ella	78	85

Using this data, test the hypothesis that the new teaching method increases students' scores. Use a 5% significance level.

(a) (2 points) State the null and alternative hypotheses.

- H_0 :
- H_a :

(b) (5 points) Calculate the test statistic.

(c) (2 points) Find (or estimate) the p -value for the test.

(d) (2 points) *State* and *interpret* your conclusion.

ANSWERS:

1. (a) 0.8222
(b) 0.3717
(c) 0.3557
(d) $P(\text{acceptable} \mid \text{Alberta}) = 0.8696$
 $P(\text{acceptable}) = 0.8222$
not equal, so events are not independent
2. (a) 0.0036
(b) 0.0830
3. (a) 0.7880
(b) 0.4513
4. (a) $P(X \geq 4) = 0.227$
(b) $\mu = 2.418$
(c) $\sigma = 1.2591$
5. (a) $P(X = 4) = 0.1586$
(b) $P(X \geq 2) = 0.7758$
6. (a) 0.8371
(b) $X = 407.5$ pounds
(c) 0.0007
7. (a) $\mu = 95.85$
(b) $\sigma = 8.6853$
(c) 0.7640
8. (a) (6.8336, 9.5664)
(b) We believe that the true average luminosity of all stars is between 6.8336 and 9.5664. The method used to construct this interval is successful at capturing the true mean 99% of the time.
9. (a) (0.8100, 0.8740)
(b) We believe that the true percentage of all adult residents in this city who possess a smartphone is between 81.00% and 87.40%. The method used to construct this interval is successful at capturing the true percentage 95% of the time.
10. sample of 35 people of age 5
11. (a) $H_0 : \mu = 8, H_a : \mu < 8$
(b) $t^* = -2.5263$
(c) $0.005 < P\text{-value} < 0.010$
(d) reject H_0
The evidence is sufficiently convincing to claim that the true average arsenic level in this well is less than 8 ppb, at the 2% significance level.
12. (a) $H_0 : p_O - p_Q = 0, H_a : p_O - p_Q > 0$
or
 $H_0 : p_Q - p_O = 0, H_a : p_Q - p_O < 0$
(b) $z^* = 0.85$ (or $z^* = -0.85$)
(c) $P\text{-value} = 0.1977$
(d) fail to reject H_0
13. (a) $H_0 : \mu_d = 0, H_a : \mu_d > 0$
(if difference = new - old)
or
 $H_0 : \mu_d = 0, H_a : \mu_d < 0$
(if difference = old - new)
(b) $t^* = 1.3804$ (or $t^* = -1.3804$)
(c) $0.100 < P\text{-value} < 0.125$
(d) fail to reject H_0
The evidence is not sufficiently convincing to claim that the new teaching method increases student's scores on average, at the 5% significance level.